

RAKIBUL HASAN SHAKIL

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RESEARCH INTERESTS

Machine Learning and Explainable AI · Fairness and Bias Mitigation in AI Systems · Deep Learning Applications in Healthcare and Medical Diagnostics · Fraud Detection and Anomaly Detection · IoT-enabled Smart Systems and Embedded Intelligence · Data Analytics and Feature Engineering

ACADEMIC BACKGROUND

- Bachelor of Science in Computer Science and Engineering**
American International University-Bangladesh (AIUB), Dhaka
Cumulative GPA: 3.78/4.00 | Honour: Cum Laude (Bronze Medal)

April 2026
- Higher Secondary Certificate (HSC)**
BAF Shaheen College, Chittagong
GPA: 5.00/5.00 (Highest Honours)

2021
- Secondary School Certificate (SSC)**
Railway Public High School
GPA: 4.89/5.00

2019

RESEARCH EXPERIENCE AND PUBLICATIONS

- Research Associate**
Smile Care Limited, Dhaka, Bangladesh

October 2025 – February 2026

 - Conducted empirical research on Machine Learning and Explainable AI (XAI) for fraud detection in large-scale credit card transaction datasets, currently under peer review for publication.
 - Developed comprehensive fairness-aware evaluation framework for mitigating demographic bias in ECG signal classification models, addressing critical challenges in medical AI deployment and clinical equity.
 - Analyzed and preprocessed large-scale clinical and transactional datasets (>100K records) utilizing data cleaning, normalization, and feature extraction techniques to ensure data quality and model reliability.
 - Collaborated with interdisciplinary research teams to design experimental protocols, validate findings, and prepare manuscripts for peer-reviewed journal submission.
 - Implemented multiple machine learning and deep learning algorithms (Random Forest, XGBoost, CNN, LSTM) with comparative performance evaluation and interpretability analysis.
- Student Assistant**
Department of Computer Science, American International University-Bangladesh

October 2025 – February 2026

 - Provided academic mentorship and administrative support to undergraduate students regarding course registration, academic requirements, and program-related inquiries.
 - Assisted faculty members with course coordination, student record management, and academic documentation.

Peer-Reviewed Publications and Conference Presentations

- **Accepted Conference Paper (Under Review for Publication):** "Development and Implementation of an Arduino-based Floor Cleaning Robot" – International Conference on Electronics and Informatics (ICEI 2026), Atomic Energy Centre, Dhaka, Bangladesh. Poster presentation scheduled for May 2026. (Co-authors: S. A. Shimu, M. Ur-Rahman, Md. Siam-Ul-Islam, T. Hasan, and Prof. Dr. Muhibul Haque Bhuyan)
- **Manuscript Under Review:** "A Comparative Analysis of Machine Learning, Deep Learning, and Explainable AI Models for Fraud Detection in Large-Scale Credit Card Transactions" – Focus on algorithm performance, interpretability trade-offs, and scalability considerations for real-world deployment.
- **Manuscript Under Review:** "Fairness-Aware Evaluation and Mitigation of Demographic Bias in ECG Classification" – Addresses critical issues of algorithmic fairness and bias detection in medical AI systems, with implications for equitable healthcare deployment.

MAJOR ACADEMIC PROJECTS

IoT-enabled Crop Recommendation System using Machine Learning

IoT Sensors | Machine Learning Algorithms | Smart Agriculture

- Designed and implemented machine learning model to predict optimal crop cultivation based on real-time environmental sensor data (temperature, humidity, soil pH, rainfall patterns).
- Engineered feature extraction and optimization pipeline, achieving improved model accuracy through systematic feature selection and dimensionality reduction techniques.
- Integrated IoT sensor deployment with cloud-based data processing pipeline for real-time decision support in precision agriculture.

Doctor HUB: Secure Patient-Doctor Communication Platform

Java | Real-Time Communication | Healthcare IT

- Architected and developed secure real-time chat module with end-to-end encryption for confidential patient-physician communication.
- Implemented automated appointment booking and scheduling system with intelligent physician availability prediction algorithm based on historical appointment patterns.
- Designed comprehensive admin dashboard for appointment management, patient record tracking, and communication analytics.

Autonomous Floor Cleaning Robot

ESP32 | Arduino | Embedded Systems | Sensor Integration

- Engineered autonomous floor-cleaning robot utilizing ESP32 microcontroller with integrated motor control via L298N drivers for brush and wheel actuation.
- Implemented real-time obstacle detection and avoidance algorithms using IR and ultrasonic sensors for autonomous navigation.
- Accepted for presentation at International Conference on Electronics and Informatics (ICEI 2026) – conference paper under review for publication.

TECHNICAL COMPETENCIES

Programming Languages:	Python, Java, C++, C#, JavaScript, PHP
Machine Learning:	TensorFlow, Scikit-Learn, XGBoost, Random Forest, Neural Networks, Deep Learning (CNN, LSTM)
Data Science:	Data Analytics, Feature Engineering, Statistical Analysis, NumPy, Pandas, Matplotlib, Data Visualization
AI & Advanced Topics:	Explainable AI (XAI), Natural Language Processing (NLP), Fairness and Bias Mitigation, Anomaly Detection

Embedded Systems: Arduino, ESP32, Sensor Integration, Real-Time Systems, Motor Control
Web Technologies: .NET Core, HTML/CSS, REST APIs, Database Management (SQL, NoSQL)

CERTIFICATIONS AND TRAINING

- NSDA Bangladesh Certification – Professional Development in Soft Skills and Communication
- Graphics Design Certification – Digital Design and UI/UX Principles

AWARDS AND ACADEMIC ACHIEVEMENTS

- **Cum Laude with Bronze Medal Distinction** – American International University-Bangladesh (Academic Honours)
- **Dean's Award for Academic Excellence** – Semester FALL 23-24, American International University-Bangladesh
- **Dean's Award for Academic Excellence** – Semester FALL 24-25, American International University-Bangladesh

LANGUAGES

English (Fluent – Professional Proficiency) · **Bangla** (Native) · **Urdu** (Basic) · **Hindi** (Basic)

ACADEMIC REFERENCES

Dr. Razuan Karim

Associate Professor, Department of Computer Science

American International University-Bangladesh

Email: rkarim@aiub.edu

Prof. Dr. Muhibul Haque Bhuyan

Professor, Department of Electrical and Electronic Engineering

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Mr. Abhijit Bhowmik

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